

E-COMMERCE SOLUTION PLAYBOOK

RAG Knowledge Base for AI Agent 3

A 30-page anomaly-to-action intervention handbook designed for solution generation in the AIC e-commerce analytics architecture.

PURPOSE

Give Agent 3 concrete, efficient, cause-matched interventions — including execution steps, guardrails, metrics and experiment logic — instead of generic business advice.

Edition: September 2026 • Project: AIC • Domain: E-commerce / Marketplace Analytics

How AI Agent 3 Should Use This Knowledge Base

This page is retrieval guidance. The remaining pages are intervention content.

INPUT FROM AGENT 2	Anomaly + diagnosed cause(s) + driver KPIs + supporting evidence / confidence.
RETRIEVAL QUERY	Encode KPI direction, confirmed cause, sub-driver and business surface. Example: “revenue down because returning customers and orders/customer declined”.
RETRIEVE	Prefer 3–6 action cards whose TRIGGER / RAG TAGS match the cause. Do not retrieve by anomaly name alone.
RANK	Filter by applicability, impact, effort, speed, confidence and evidence available in the business data.
GENERATE	Return 1 primary action, 2–4 supporting actions, “do not prioritize” guidance, metrics, and validation plan.

Core decision principles

- 1. Cause first: recommend against the cause identified by Agent 2, not merely the top-level KPI anomaly.
- 2. Fix → Optimize → Grow: repair leakage (availability, payment, checkout, service) before adding aggressive demand.
- 3. Prefer targeted interventions over site-wide actions. Segmentation generally increases efficiency and reduces cannibalization.
- 4. Separate observed facts from hypotheses. If required evidence is absent, frame the action as an investigation or controlled test.
- 5. Protect guardrails: margin, conversion, cancellation, return rate, customer experience and seller quality may matter as much as the target KPI.
- 6. High-cost or structural changes require validation. Default to an experiment, phased rollout, holdout or before/after cohort design.

Recommended Agent 3 answer shape

Problem → Confirmed cause → Recommended strategy → Primary action → Supporting actions → Do not prioritize → Expected mechanism → Metrics to monitor → Experiment / validation.

Customer Acquisition Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

01. Reallocate Acquisition Toward Customer Quality, Not Raw Volume

RAG TAGS: revenue_down | new_customers_down | acquisition | channel_mix | high_efficiency

TRIGGER: New-customer count is falling, or acquisition is producing weak downstream value while retention and basket metrics are not the main problem.

RECOMMENDED ACTION: Shift acquisition effort toward channels, audiences, geographies and products that generate qualified first purchases and stronger repeat behavior rather than optimizing only for clicks or top-line order count.

EXECUTION: Rank acquisition sources by first-order conversion, early repeat rate, AOV and cancellation/return quality. Reduce spend on low-quality sources, protect high-performing cohorts, and rebuild campaign/landing-page messages around the products and customer segments that create durable value.

USE WHEN: Source or campaign attribution exists, and the business can distinguish new-customer origin or at least geography/category of acquisition.

AVOID / DE-PRIORITIZE WHEN: The confirmed cause is returning-customer loss, checkout failure, seller supply loss, or fulfillment deterioration. Do not infer a channel problem without source-level evidence.

MONITOR: New customers, first-order conversion, repeat purchase rate by source, AOV by source, cancellation/return rate, incremental revenue.

EFFICIENCY: Impact: High | Effort: Medium | Speed: Medium | Confidence: High when channel-level data is available.

EXPERIMENT / VALIDATION: Move a controlled share of budget from weak to strong segments and compare incremental new customers plus 30/60-day quality metrics, not just clicks.

EVIDENCE ANCHORS: [R1] [R12]

02. Use a Controlled First-Purchase Offer

RAG TAGS: new_customers_down | conversion | promotion | first_order | margin_guardrail

TRIGGER: New-customer conversion or first-order volume is weak and the business needs a short-term acquisition lever.

RECOMMENDED ACTION: Offer a targeted first-purchase incentive with a clear eligibility rule and an explicit margin ceiling, rather than applying a site-wide discount to every visitor.

EXECUTION: Choose one acquisition segment, set a minimum basket or product eligibility rule, cap the discount, exclude low-margin SKUs if needed, and suppress the offer for returning customers who would purchase anyway. Pair it with a relevant landing page and a post-purchase path to purchase #2.

USE WHEN: Price sensitivity is plausible, margin is understood, and the offer can be targeted to true first-time buyers.

AVOID / DE-PRIORITIZE WHEN: Average item price is already deteriorating because of over-discounting, or the primary issue is poor availability, weak retention, or checkout friction.

MONITOR: First-order conversion, new customers, contribution margin/order, AOV, second-purchase rate, promo redemption.

EFFICIENCY: Impact: Medium-High | Effort: Low | Speed: Fast | Confidence: Medium unless tested.

EXPERIMENT / VALIDATION: A/B test offer vs no offer for qualified new visitors; require incremental conversion and acceptable contribution margin before scaling.

EVIDENCE ANCHORS: [R1] [R12]

Customer Acquisition Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

03. Launch a Customer Referral Loop

RAG TAGS: new_customers_down | acquisition | referral | retention_synergy | organic_growth

TRIGGER: New-customer growth is weak but the business has a meaningful base of satisfied repeat customers.

RECOMMENDED ACTION: Create a referral program that rewards an existing customer only when a referred customer completes a qualified first purchase.

EXECUTION: Define a referral reward, minimum order criteria, fraud rules and expiry. Target high-satisfaction or repeat cohorts first, provide simple sharing links/codes, and attach the referral prompt to post-purchase or loyalty communications rather than interrupting checkout.

USE WHEN: There is an identifiable repeat-customer base and sufficient product satisfaction or advocacy potential.

AVOID / DE-PRIORITIZE WHEN: Customer service, returns, delivery reliability or product quality are currently poor; referral incentives amplify a weak experience rather than fixing it.

MONITOR: Referred customers, referral conversion, cost per referred customer, repeat rate of referred customers, fraud rate.

EFFICIENCY: Impact: Medium | Effort: Medium | Speed: Medium | Confidence: Medium.

EXPERIMENT / VALIDATION: Pilot with a high-value customer cohort and compare referred-customer quality with paid-acquisition cohorts.

EVIDENCE ANCHORS: [R6] [R12]

04. Focus Acquisition on High-Response Categories and Geographies

RAG TAGS: new_customers_down | geography | category | segmentation | resource_allocation

TRIGGER: New-customer decline is concentrated in certain regions or categories while other segments remain healthy.

RECOMMENDED ACTION: Concentrate acquisition resources where demand, conversion and supply coverage are strongest; use weak segments for diagnosis rather than spreading spend evenly.

EXECUTION: Create a region x category matrix of new customers, conversion proxies, AOV, seller/assortment coverage and growth. Scale strong cells, repair strategically important weak cells, and avoid purchasing traffic into areas with poor availability or service coverage.

USE WHEN: The dataset can segment by geography or category and the decline is not uniform.

AVOID / DE-PRIORITIZE WHEN: The anomaly is broad-based with no segment concentration, or supply/fulfillment constraints would make extra demand harmful.

MONITOR: New customers by region/category, conversion, AOV, seller coverage, cancellations, delivery performance.

EFFICIENCY: Impact: High | Effort: Medium | Speed: Medium | Confidence: High when segment concentration is clear.

EXPERIMENT / VALIDATION: Run geo/category holdouts or phased budget changes and compare incremental customer growth with service guardrails.

EVIDENCE ANCHORS: [R1] [R12]

Customer Acquisition Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

05. Improve Product-Feed and Search Discoverability

RAG TAGS: acquisition | product_discovery | listing_quality | search | merchant_feed

TRIGGER: Customer acquisition is weak for products with healthy availability and competitive commercial conditions, suggesting discoverability or listing relevance may be limiting reach.

RECOMMENDED ACTION: Improve titles, attributes, images, identifiers, category mapping, price/availability freshness and feed-to-landing-page consistency for priority products.

EXECUTION: Prioritize high-value/high-demand SKUs. Put important attributes early in titles, use high-quality images, complete product attributes and variants, keep price and availability current, and resolve mismatches between feed and landing page.

USE WHEN: The business relies on product search, shopping feeds, marketplaces or catalog discovery and has product-level performance data.

AVOID / DE-PRIORITIZE WHEN: The primary issue is no demand, poor seller supply, severe delivery problems or uncompetitive economics; listing optimization cannot repair a structural offer problem.

MONITOR: Impressions/search visibility where available, product clicks, product-page visits, conversion, disapprovals/feed errors, revenue from optimized SKUs.

EFFICIENCY: Impact: Medium-High | Effort: Medium | Speed: Medium | Confidence: High for catalog-quality defects.

EXPERIMENT / VALIDATION: Optimize a matched set of priority SKUs and compare search/discovery and conversion changes against similar unchanged SKUs.

EVIDENCE ANCHORS: [R1] [R2] [R10]

06. Repair Landing-Page Message Match

RAG TAGS: acquisition | conversion | landing_page | message_match | product_page

TRIGGER: Traffic or acquisition volume is present but product-page conversion is weak or visitors land on pages that do not clearly match the promoted product/variant.

RECOMMENDED ACTION: Make the landing page immediately confirm the product, variant, price, availability and purchase action promised upstream.

EXECUTION: Send users to the exact product detail page, preselect the promoted variant, surface price and availability clearly, remove obstructive pop-ups, preserve the same offer/attributes across devices, and keep the buy/add-to-cart action visible.

USE WHEN: The acquisition path includes ads, feeds, marketplaces, email or campaign landing pages and there is evidence of product-page drop-off or mismatch.

AVOID / DE-PRIORITIZE WHEN: The main problem occurs after add-to-cart, or the advertised product is genuinely unavailable; fix availability/checkout first.

MONITOR: Landing-page bounce/drop-off proxy, add-to-cart rate, conversion, variant errors, price/availability mismatch incidents.

EFFICIENCY: Impact: High | Effort: Low-Medium | Speed: Fast-Medium | Confidence: High when mismatch exists.

EXPERIMENT / VALIDATION: Test corrected product-page routing/content against current landing flow for matched traffic.

EVIDENCE ANCHORS: [R2] [R1]

Customer Retention & Reactivation Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

07. Run 30/60/90-Day Win-Back Segments

RAG TAGS: returning_customers_down | retention | winback | reactivation | revenue_down

TRIGGER: Returning-customer count is falling and the decline is driven by previously active customers becoming inactive.

RECOMMENDED ACTION: Segment lapsed customers by recency and prior value, then apply progressively stronger reactivation treatments instead of a single broad win-back message.

EXECUTION: Create 30/60/90-day inactive cohorts adjusted for normal purchase cycle. Start with useful reminders or relevant products; reserve stronger incentives for later/high-value lapse segments. Suppress customers who have already returned.

USE WHEN: Customer-level order history exists and the business can distinguish new vs returning customers.

AVOID / DE-PRIORITIZE WHEN: The category naturally has very long purchase cycles, or customers are lapsing because of unresolved delivery/service issues. Fix structural experience failures first.

MONITOR: Reactivation rate, returning customers, incremental repeat orders, orders/customer, contribution margin from win-back.

EFFICIENCY: Impact: High | Effort: Low-Medium | Speed: Fast | Confidence: High for genuine lapse cohorts.

EXPERIMENT / VALIDATION: Hold out part of each lapse cohort; compare incremental return rate and margin across reminder-only vs incentive treatments.

EVIDENCE ANCHORS: [R5] [R6] [R12]

08. Accelerate the Second Purchase

RAG TAGS: retention | new_to_repeat | second_purchase | customer_lifecycle | orders_per_customer

TRIGGER: New-customer acquisition is healthy but returning-customer growth or repeat rate is weak.

RECOMMENDED ACTION: Treat purchase #2 as a dedicated conversion event and build a short post-first-order journey designed to move customers into repeat behavior.

EXECUTION: After delivery, send product-use guidance, complementary recommendations, replenishment timing where relevant, and a time-bounded second-purchase benefit only if economics justify it. Personalize by first category or SKU.

USE WHEN: The business has many one-time buyers and can identify first-order date/category.

AVOID / DE-PRIORITIZE WHEN: Product quality, fulfillment or returns are deteriorating; pushing purchase #2 before repairing experience can reduce trust.

MONITOR: Second-purchase rate, days to second order, returning customers, orders/customer, margin on second orders.

EFFICIENCY: Impact: High | Effort: Medium | Speed: Medium | Confidence: High when one-time-buyer share is high.

EXPERIMENT / VALIDATION: Randomize first-time buyers into standard vs accelerated lifecycle treatment and measure second-order lift.

EVIDENCE ANCHORS: [R5] [R6] [R12]

Customer Retention & Reactivation Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

09. Build a Post-Purchase Retention Sequence

RAG TAGS: retention | post_purchase | customer_experience | repeat_purchase | crm

TRIGGER: Repeat purchasing is weakening after otherwise successful first orders.

RECOMMENDED ACTION: Use the period after purchase to reinforce product value, reduce uncertainty and create the next relevant purchase opportunity.

EXECUTION: Sequence order confirmation, delivery/use guidance, support access, review request, complementary recommendation and later reorder/win-back communication. Keep messages useful and suppress irrelevant promotions while an order issue is unresolved.

USE WHEN: Email/SMS/app communication is available and customer consent/permissions allow lifecycle messaging.

AVOID / DE-PRIORITIZE WHEN: Do not over-message low-frequency categories or send promotional flows during active complaints, refunds or failed deliveries.

MONITOR: Repeat purchase rate, returning customers, email/SMS engagement, support contact rate, unsubscribe rate, incremental revenue.

EFFICIENCY: Impact: Medium-High | Effort: Medium | Speed: Medium | Confidence: Medium-High.

EXPERIMENT / VALIDATION: Test lifecycle cadence and content by first-purchase category with holdout cohorts.

EVIDENCE ANCHORS: [R5] [R6]

10. Trigger Replenishment Reminders from Expected Reorder Cycles

RAG TAGS: orders_per_customer_down | retention | replenishment | frequency | crm

TRIGGER: Orders per customer are falling in categories that customers reasonably repurchase or replace.

RECOMMENDED ACTION: Estimate the normal reorder interval and contact customers near the expected replenishment window with a direct path to reorder.

EXECUTION: Calculate median/typical reorder days by SKU/category or customer segment. Send reminders shortly before/after that window, preload prior products where possible, and avoid blanket weekly messaging.

USE WHEN: Products are consumable, replaceable or repeatedly purchased and historical repeat orders exist.

AVOID / DE-PRIORITIZE WHEN: Durable one-time-purchase categories, uncertain purchase cycles, or customers with unresolved service/return issues.

MONITOR: Reorder conversion, days between orders, orders/customer, returning customers, unsubscribe rate.

EFFICIENCY: Impact: High for replenishable categories | Effort: Medium | Speed: Medium | Confidence: High when cycle is stable.

EXPERIMENT / VALIDATION: Compare timing windows and reminder-only vs reminder-plus-incentive; optimize for incremental repeat orders.

EVIDENCE ANCHORS: [R5] [R6] [R12]

Customer Retention & Reactivation Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

11. Use Loyalty Rewards to Reinforce Repeat Behavior

RAG TAGS: retention | loyalty | repeat_purchase | customer_value | frequency

TRIGGER: Returning customers or purchase frequency are weak despite acceptable acquisition and service performance.

RECOMMENDED ACTION: Create rewards that encourage valuable repeat behavior rather than indiscriminate discounting.

EXECUTION: Reward repeat purchases, referrals, category milestones or spend tiers. Keep rules simple, make progress visible, and design the value so customers have a reason to return without permanently lowering price.

USE WHEN: There is a meaningful repeat-purchase opportunity and customer identity can be tracked across orders.

AVOID / DE-PRIORITIZE WHEN: The business has low natural repeatability, weak margins, or major customer-experience defects; loyalty cannot compensate for poor fundamentals.

MONITOR: Repeat purchase rate, member vs non-member frequency, redemption rate, AOV, contribution margin, customer retention.

EFFICIENCY: Impact: Medium-High | Effort: Medium-High | Speed: Medium-Long | Confidence: Medium.

EXPERIMENT / VALIDATION: Pilot a simple reward mechanic in a defined cohort/category before launching a complex tiered program.

EVIDENCE ANCHORS: [R6] [R8] [R12]

12. Personalize Recommendations from Purchase History

RAG TAGS: retention | cross_sell | personalization | repeat_purchase | customer_history

TRIGGER: Customers are returning less often or buying narrowly despite a broad catalog of relevant complementary products.

RECOMMENDED ACTION: Use previous purchases and category affinity to recommend the next most relevant product instead of showing the same generic products to everyone.

EXECUTION: Start with deterministic rules such as complementary categories, repeat purchase history and popularity within similar buyers. Exclude already-owned durable substitutes where inappropriate and control recommendation frequency.

USE WHEN: Customer-level purchase history and product/category relationships are available.

AVOID / DE-PRIORITIZE WHEN: Sparse history, tiny catalogs, or weak product relevance. Do not claim personalization if the recommendation is not based on customer behavior.

MONITOR: Recommendation click/add-to-cart, repeat orders, cross-category penetration, AOV, orders/customer.

EFFICIENCY: Impact: Medium-High | Effort: Medium | Speed: Medium | Confidence: Medium-High.

EXPERIMENT / VALIDATION: Compare personalized vs generic recommendations using incremental orders/AOV, not only clicks.

EVIDENCE ANCHORS: [R6] [R8] [R12]

Customer Retention & Reactivation Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

13. Prioritize High-Value At-Risk Customers

RAG TAGS: retention | high_value | churn_risk | reactivation | resource_efficiency

TRIGGER: Returning-customer decline is concentrated among historically valuable or frequent buyers.

RECOMMENDED ACTION: Use retention resources first on customers with high prior value and credible reactivation potential.

EXECUTION: Score customers using recency, frequency, monetary value, prior margin where available and service status. Route the highest-value at-risk cohorts to personalized offers, service outreach or concierge support; automate lower-value cohorts.

USE WHEN: Customer history can distinguish high-value from low-value customers.

AVOID / DE-PRIORITIZE WHEN: Do not use spend history alone if high-value customers are also high-return/high-support-cost. Include quality/economic guardrails where possible.

MONITOR: Reactivation by value tier, retained revenue, margin, contact cost, repeat rate.

EFFICIENCY: Impact: High | Effort: Medium | Speed: Fast-Medium | Confidence: High when value concentration is strong.

EXPERIMENT / VALIDATION: Use matched holdouts inside value tiers to estimate incremental retained revenue per intervention cost.

EVIDENCE ANCHORS: [R5] [R6]

14. Use Service Recovery as a Retention Intervention

RAG TAGS: returning_customers_down | service | delivery | returns | retention

TRIGGER: Retention decline overlaps with late deliveries, cancellations, returns, complaints or seller-service deterioration.

RECOMMENDED ACTION: Fix the experience failure and proactively recover affected customers before spending on generic loyalty or acquisition.

EXECUTION: Identify affected orders/customers, explain the issue clearly, resolve refunds/replacements quickly, offer service recovery proportional to failure severity, and feed root causes back to seller/fulfillment operations.

USE WHEN: There is observable evidence connecting churn/retention decline to operational failures.

AVOID / DE-PRIORITIZE WHEN: Do not compensate every inactive customer without evidence of service failure; that can create unnecessary cost and hide the real cause.

MONITOR: Repeat rate after recovery, complaint resolution time, refund/replacement cycle, returning customers, cancellation/late-delivery rates.

EFFICIENCY: Impact: High | Effort: Medium | Speed: Medium | Confidence: High when issue linkage is explicit.

EXPERIMENT / VALIDATION: Track recovered customers against comparable affected customers under the prior service process.

EVIDENCE ANCHORS: [R6] [R11]

Conversion & Checkout Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

15. Enable and Clearly Surface Guest Checkout

RAG TAGS: conversion_down | checkout | guest_checkout | cart_abandonment | quick_win

TRIGGER: Checkout completion is weak and account creation is mandatory or the guest path is hidden.

RECOMMENDED ACTION: Allow customers to purchase without creating an account and move optional account creation to after the order.

EXECUTION: Place guest checkout prominently, minimize sign-in pressure, preserve order tracking through email/phone, and offer account creation on confirmation using information already entered.

USE WHEN: Customers currently face forced account creation or confusing sign-in choices.

AVOID / DE-PRIORITIZE WHEN: Regulatory or product requirements genuinely require identity/account verification; in those cases minimize the required step rather than bypass it.

MONITOR: Checkout completion, abandonment at sign-in step, new-customer conversion, account creation after purchase.

EFFICIENCY: Impact: High | Effort: Low-Medium | Speed: Fast | Confidence: High when forced registration exists.

EXPERIMENT / VALIDATION: A/B test guest-first presentation vs current sign-in flow.

EVIDENCE ANCHORS: [R3] [R4]

16. Remove Nonessential Checkout Fields and Steps

RAG TAGS: conversion_down | checkout | friction | form_optimization | mobile

TRIGGER: Users reach checkout but completion falls through long forms, unnecessary steps or repeated data entry.

RECOMMENDED ACTION: Reduce checkout to the minimum information needed to complete payment and fulfillment.

EXECUTION: Audit every field, remove optional questions, use sensible defaults/autocomplete, avoid re-asking known data, combine steps where clarity improves, and make validation errors immediate and specific.

USE WHEN: Checkout has avoidable fields/steps or a high drop-off at form entry.

AVOID / DE-PRIORITIZE WHEN: Do not remove legally required or fulfillment-critical information; simplify collection rather than creating downstream delivery risk.

MONITOR: Checkout step completion, form error rate, completion time, conversion, support contacts.

EFFICIENCY: Impact: High | Effort: Medium | Speed: Medium | Confidence: High when friction is visible.

EXPERIMENT / VALIDATION: Test simplified flow with identical commercial terms and measure completed purchases plus error/fulfillment guardrails.

EVIDENCE ANCHORS: [R3] [R4]

Conversion & Checkout Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

17. Show Shipping, Taxes and Fees Before the Final Step

RAG TAGS: conversion_down | checkout | shipping_cost | fee_transparency | cart_abandonment

TRIGGER: Checkout abandonment is high and customers encounter delivery charges, taxes or other fees late in the journey.

RECOMMENDED ACTION: Expose total cost—or a clear estimate—earlier so customers can make an informed decision before investing effort in checkout.

EXECUTION: Show shipping policy and threshold in cart, estimate taxes/fees when possible, label unavoidable uncertainty, and keep total cost updated as items/addresses change.

USE WHEN: Late cost surprise is plausible or observed in user research/support/cart data.

AVOID / DE-PRIORITIZE WHEN: Do not promise free/low shipping if fulfillment economics cannot support it; transparency is better than a misleading headline.

MONITOR: Cart-to-checkout, checkout completion, abandonment after shipping step, AOV, shipping margin.

EFFICIENCY: Impact: High | Effort: Low-Medium | Speed: Fast | Confidence: High for late-cost surprises.

EXPERIMENT / VALIDATION: Compare earlier cost disclosure vs current flow while holding prices constant.

EVIDENCE ANCHORS: [R3] [R7]

18. Expand Relevant Payment Methods and Recover Payment Failures

RAG TAGS: conversion_down | payments | checkout | payment_failure | recovery

TRIGGER: Customers reach payment but fail or abandon because preferred methods are missing or transactions error out.

RECOMMENDED ACTION: Support payment methods that match the market and add a clear recovery path for failed transactions.

EXECUTION: Analyze failure codes and method usage, prioritize high-demand local methods, preserve cart/order state after failure, let customers retry or switch method without re-entering data, and communicate failure clearly.

USE WHEN: Payment-stage drop-off or failure data exists.

AVOID / DE-PRIORITIZE WHEN: Do not add every possible payment method without demand, operational readiness or fraud controls.

MONITOR: Payment success rate, failure rate by method, retry success, checkout completion, fraud/chargeback guardrails.

EFFICIENCY: Impact: High | Effort: Medium-High | Speed: Medium | Confidence: High when payment failures are material.

EXPERIMENT / VALIDATION: Roll out new/recovery methods to a controlled traffic share and compare successful orders plus fraud/cost.

EVIDENCE ANCHORS: [R4]

Conversion & Checkout Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

19. Optimize Checkout for Mobile Completion

RAG TAGS: conversion_down | mobile | checkout | ux | payment

TRIGGER: Mobile traffic is substantial and mobile checkout converts materially worse than desktop or shows more form/payment friction.

RECOMMENDED ACTION: Design the checkout around small-screen completion rather than shrinking the desktop flow.

EXECUTION: Use mobile-friendly fields, address/payment autofill, large tap targets, minimal typing, persistent progress/order summary, fast page performance and relevant mobile wallets where appropriate.

USE WHEN: Device-level data shows a mobile gap or usability review reveals mobile-specific friction.

AVOID / DE-PRIORITIZE WHEN: Do not assume mobile is the problem from traffic share alone; require a conversion or UX gap.

MONITOR: Mobile checkout completion, time-to-complete, field error rate, payment success, device conversion gap.

EFFICIENCY: Impact: High | Effort: Medium | Speed: Medium | Confidence: High with a measured mobile gap.

EXPERIMENT / VALIDATION: Run mobile-specific A/B tests and keep desktop unchanged to isolate impact.

EVIDENCE ANCHORS: [R4] [R3]

20. Recover Abandoned Carts Without Training Discount Dependence

RAG TAGS: conversion_down | cart_abandonment | crm | recovery | discount_guardrail

TRIGGER: Customers add products to cart but do not complete purchase.

RECOMMENDED ACTION: Use a staged cart-recovery sequence that begins with convenience and relevance, not an automatic discount.

EXECUTION: Send a reminder with cart contents, product availability and direct return-to-cart link. Add support/trust information where useful. Use incentives only for selected high-propensity or late-stage segments after testing incremental value.

USE WHEN: Customer identity/consent exists and cart persistence is reliable.

AVOID / DE-PRIORITIZE WHEN: Do not send repeated recovery messages to customers who already purchased, removed the cart intentionally, or have unresolved service issues.

MONITOR: Recovered orders, incremental conversion vs holdout, margin, unsubscribe rate, time to recovery.

EFFICIENCY: Impact: Medium-High | Effort: Low | Speed: Fast | Confidence: High with holdout measurement.

EXPERIMENT / VALIDATION: Hold out a portion of abandoned carts and compare reminder-only, social-proof/support, and incentive variants.

EVIDENCE ANCHORS: [R3] [R12]

Conversion & Checkout Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

21. Add Credible Product Reviews and Social Proof

RAG TAGS: conversion_down | reviews | trust | product_page | acquisition_quality

TRIGGER: Product pages receive traffic but customers lack confidence, especially for unfamiliar products or sellers.

RECOMMENDED ACTION: Collect and surface authentic, product-specific reviews and quality signals that help customers evaluate purchase risk.

EXECUTION: Request reviews after verified delivery, preserve negative as well as positive feedback, match reviews to the correct product/variant, summarize useful themes, and avoid fabricated or low-quality boilerplate.

USE WHEN: Products have low trust/consideration and the business can collect legitimate customer feedback.

AVOID / DE-PRIORITIZE WHEN: Do not treat reviews as a substitute for fixing bad products, misleading listings or unreliable fulfillment.

MONITOR: Review coverage, product-page conversion, return rate, rating distribution, review helpfulness/quality.

EFFICIENCY: Impact: Medium | Effort: Medium | Speed: Medium-Long | Confidence: Medium-High.

EXPERIMENT / VALIDATION: Compare review-rich vs low-review products carefully using matched cohorts or staged review collection.

EVIDENCE ANCHORS: [R9] [R13]

22. Strengthen Product-Detail Clarity Before Adding More Traffic

RAG TAGS: conversion_down | product_page | listing_quality | trust | content

TRIGGER: Acquisition traffic is adequate but add-to-cart or product conversion is weak because the offer is difficult to evaluate.

RECOMMENDED ACTION: Improve the product detail page so customers can quickly understand what the product is, which variant they are buying, why it is useful, what it costs and when it will arrive.

EXECUTION: Clarify title, images, key attributes, variant choices, price, availability, shipping/returns, seller information and primary CTA. Use comparison content when customers need help choosing between variants or related products.

USE WHEN: The product page is incomplete, ambiguous or inconsistent with feed/listing information.

AVOID / DE-PRIORITIZE WHEN: Do not overload simple products with excessive content; clarity beats quantity.

MONITOR: Product-page conversion, add-to-cart rate, variant errors, return reasons, customer questions.

EFFICIENCY: Impact: Medium-High | Effort: Medium | Speed: Medium | Confidence: High for obvious content gaps.

EXPERIMENT / VALIDATION: Test focused content changes on priority SKUs rather than redesigning the whole catalog at once.

EVIDENCE ANCHORS: [R1] [R10]

Purchase Frequency Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

23. Make Reordering Frictionless

RAG TAGS: orders_per_customer_down | frequency | reorder | retention | convenience

TRIGGER: Customers still return but order frequency is falling, especially for repeatable products.

RECOMMENDED ACTION: Reduce the effort required to buy a previously purchased product again.

EXECUTION: Provide order-history reorder links, prefilled carts where appropriate, saved preferences, simple quantity changes and clear stock/substitute information. Preserve customer control before payment.

USE WHEN: Repeat purchase behavior exists and products are commonly reordered.

AVOID / DE-PRIORITIZE WHEN: Do not push one-click reorder for products with high substitution, customization or safety requirements without confirmation.

MONITOR: Reorder conversion, orders/customer, reorder time, basket changes, cancellation/return rate.

EFFICIENCY: Impact: Medium-High | Effort: Medium | Speed: Medium | Confidence: High in repeatable categories.

EXPERIMENT / VALIDATION: Expose reorder shortcuts to a cohort of previous buyers and compare repeat frequency.

EVIDENCE ANCHORS: [R6] [R4]

24. Use Category-Triggered CRM Instead of Calendar-Only Blasts

RAG TAGS: orders_per_customer_down | crm | category | personalization | frequency

TRIGGER: Purchase frequency is declining and broad campaign calendars generate low relevance.

RECOMMENDED ACTION: Trigger communication from customer category behavior, purchase stage or replenishment logic rather than sending identical promotions to the whole base.

EXECUTION: Define category-entry, cross-category, replenishment and lapse triggers. Choose one objective per message, suppress irrelevant segments and cap frequency to avoid fatigue.

USE WHEN: Customer-category history is available and messaging channels are permissioned.

AVOID / DE-PRIORITIZE WHEN: Do not create hundreds of fragile micro-segments before basic lifecycle measurement works.

MONITOR: Orders/customer, incremental conversion, message fatigue/unsubscribes, cross-category penetration.

EFFICIENCY: Impact: Medium-High | Effort: Medium | Speed: Medium | Confidence: Medium-High.

EXPERIMENT / VALIDATION: Compare behavior-triggered campaigns with standard calendar campaigns using holdout groups.

EVIDENCE ANCHORS: [R5] [R12]

Purchase Frequency Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

25. Offer Subscription or Auto-Replenishment Selectively

RAG TAGS: orders_per_customer_down | subscription | replenishment | frequency | retention

TRIGGER: Purchase frequency is weak in products with predictable, recurring consumption.

RECOMMENDED ACTION: Offer optional scheduled replenishment or subscription where convenience creates genuine customer value.

EXECUTION: Identify stable repeat intervals, offer flexible frequency/skip/cancel controls, make pricing transparent, and avoid locking customers into unsuitable quantities.

USE WHEN: The category has recurring consumption and a repeat-order pattern strong enough to justify automation.

AVOID / DE-PRIORITIZE WHEN: Durable or irregularly purchased products, volatile demand, high return rates, or weak supply reliability.

MONITOR: Subscription adoption, retention, skip/cancel rate, orders/customer, fulfillment failures, margin.

EFFICIENCY: Impact: High in suitable categories | Effort: High | Speed: Medium-Long | Confidence: Medium-High.

EXPERIMENT / VALIDATION: Pilot on one replenishable category with an opt-in cohort before wider rollout.

EVIDENCE ANCHORS: [R6] [R12]

26. Create a Lapse-and-Season Reactivation Calendar

RAG TAGS: orders_per_customer_down | seasonality | reactivation | crm | frequency

TRIGGER: Orders/customer decline follows seasonal or category-specific buying cycles rather than permanent churn.

RECOMMENDED ACTION: Schedule reactivation around expected need states and seasonal demand windows instead of treating all inactivity as churn.

EXECUTION: Estimate normal interval and seasonal peaks by category, create pre-season reminders or replenishment prompts, and separate seasonal inactivity from true retention failure.

USE WHEN: Historical data shows repeatable seasonal or periodic demand patterns.

AVOID / DE-PRIORITIZE WHEN: Do not force seasonal logic onto nonseasonal categories or use it to excuse structural retention decline.

MONITOR: Seasonal reactivation, orders/customer, timing of repeat purchase, incremental revenue vs prior season.

EFFICIENCY: Impact: Medium | Effort: Low-Medium | Speed: Medium | Confidence: High with clear seasonality.

EXPERIMENT / VALIDATION: Use staggered send windows or holdouts to find the best lead time.

EVIDENCE ANCHORS: [R5] [R12]

Purchase Frequency Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

27. Activate the Next Relevant Category

RAG TAGS: orders_per_customer_down | cross_category | frequency | personalization | growth

TRIGGER: Customers make a first or occasional purchase but do not explore adjacent categories despite a broad relevant assortment.

RECOMMENDED ACTION: Use the first purchase to identify the most plausible next category and create a focused cross-category journey.

EXECUTION: Build simple product/category adjacency rules from co-purchase or sequence data. Recommend one or two relevant categories, use educational content where needed, and avoid presenting dozens of unrelated options.

USE WHEN: Cross-category relationships exist and assortment coverage is reliable.

AVOID / DE-PRIORITIZE WHEN: If items/order is already high but frequency is low because purchase cycles are long; do not manufacture unnecessary frequency.

MONITOR: Cross-category penetration, second-category conversion, orders/customer, AOV, repeat rate.

EFFICIENCY: Impact: Medium | Effort: Medium | Speed: Medium | Confidence: Medium.

EXPERIMENT / VALIDATION: Compare adjacency-based recommendations with generic bestsellers.

EVIDENCE ANCHORS: [R6] [R8] [R12]

28. Use Frequently-Bought-Together Cross-Sells

RAG TAGS: aov_down | items_per_order_down | cross_sell | basket | quick_win

TRIGGER: AOV is falling because items per order are declining while average item price is relatively stable.

RECOMMENDED ACTION: Recommend a small set of genuinely complementary products at product page or cart stage.

EXECUTION: Mine co-purchase pairs or use domain rules, rank by relevance and attach rate, limit to a few choices, and avoid replacing the primary product decision with unrelated upsells.

USE WHEN: Complementary products exist and basket decline—not conversion—is the main issue.

AVOID / DE-PRIORITIZE WHEN: Checkout is already confusing, conversion is sharply down, or recommendations are low relevance.

MONITOR: Items/order, attach rate, AOV, conversion, incremental margin.

EFFICIENCY: Impact: High | Effort: Low-Medium | Speed: Fast-Medium | Confidence: High with strong pair affinity.

EXPERIMENT / VALIDATION: A/B test recommendation placement and pair logic; require AOV/margin lift without conversion loss.

EVIDENCE ANCHORS: [R8] [R12]

Basket & Average Order Value Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

29. Create Product Bundles Around Real Use Cases

RAG TAGS: aov_down | items_per_order_down | bundles | basket | merchandising

TRIGGER: Customers often need several complementary products but purchase them separately or omit accessories/add-ons.

RECOMMENDED ACTION: Package complementary products into simple use-case bundles with a clear combined value proposition.

EXECUTION: Use co-purchase data and merchandising judgment to create bundles, set bundle economics, ensure components are available, explain the use case, and allow sensible variant choices.

USE WHEN: Items/order is weak and bundle components have reliable availability/margin.

AVOID / DE-PRIORITIZE WHEN: Do not bundle unrelated slow movers simply to clear stock, or create discount-heavy bundles that destroy margin.

MONITOR: Bundle conversion, items/order, AOV, margin/order, component stockouts.

EFFICIENCY: Impact: High | Effort: Medium | Speed: Medium | Confidence: Medium-High.

EXPERIMENT / VALIDATION: Test bundle vs separate-product presentation on matched traffic and track margin as a guardrail.

EVIDENCE ANCHORS: [R8] [R12]

30. Set a Free-Shipping Threshold Just Above Typical Basket Value

RAG TAGS: aov_down | items_per_order_down | shipping_threshold | basket | promotion

TRIGGER: AOV/items per order are weak and shipping economics allow an incentive for slightly larger baskets.

RECOMMENDED ACTION: Set a free-shipping threshold above the typical/median basket so customers have a reason to add one useful item.

EXECUTION: Estimate median and average order value, model shipping cost and margin, choose a threshold that is attainable rather than extreme, and show progress toward the threshold in cart.

USE WHEN: Shipping cost is a meaningful purchase factor and incremental basket margin can fund the incentive.

AVOID / DE-PRIORITIZE WHEN: Thin margins, very high shipping cost, low availability, or a conversion problem caused by other checkout friction.

MONITOR: AOV, items/order, threshold attainment, conversion, shipping cost/order, contribution margin.

EFFICIENCY: Impact: Medium-High | Effort: Low | Speed: Fast | Confidence: High when unit economics are modeled.

EXPERIMENT / VALIDATION: Test threshold levels; choose the one that improves contribution margin, not merely AOV.

EVIDENCE ANCHORS: [R7] [R3]

Basket & Average Order Value Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

31. Use Quantity Packs and Multi-Buy Offers for Repeatable SKUs

RAG TAGS: aov_down | items_per_order_down | multi_buy | quantity_pack | basket

TRIGGER: Customers buy repeatable products one unit at a time despite predictable future need.

RECOMMENDED ACTION: Offer multi-unit packs or modest multi-buy economics that increase useful quantity without creating waste.

EXECUTION: Select high-repeat, low-return SKUs; create 2-pack/3-pack options, show unit economics transparently, and ensure packaging/fulfillment can support larger quantities.

USE WHEN: Products are consumable or commonly bought in multiples and inventory is reliable.

AVOID / DE-PRIORITIZE WHEN: Perishable products, uncertain fit, high return categories or items where excess quantity harms customer value.

MONITOR: Units/order, AOV, margin/unit and margin/order, return/cancel rate, repeat timing.

EFFICIENCY: Impact: Medium | Effort: Low-Medium | Speed: Fast-Medium | Confidence: Medium-High.

EXPERIMENT / VALIDATION: Test quantity pack vs single-unit default while tracking margin and customer satisfaction.

EVIDENCE ANCHORS: [R8] [R12]

32. Introduce Good-Better-Best Upsell Paths

RAG TAGS: aov_down | average_item_price_down | upsell | premium_mix | pricing

TRIGGER: AOV is falling because average item price or premium mix is declining while items per order is stable.

RECOMMENDED ACTION: Make higher-value alternatives easy to compare so customers can choose a better fit rather than forcing a blanket price increase.

EXECUTION: Group comparable products into good/better/best tiers, clarify incremental benefits, surface the upgrade near the decision point, and protect the entry option for price-sensitive customers.

USE WHEN: The catalog contains meaningful quality/feature tiers and customers can understand the value difference.

AVOID / DE-PRIORITIZE WHEN: Do not upsell when premium inventory is unreliable, differences are cosmetic, or the core conversion problem is price sensitivity.

MONITOR: Average item price, premium mix, AOV, conversion, return rate.

EFFICIENCY: Impact: Medium-High | Effort: Medium | Speed: Medium | Confidence: Medium.

EXPERIMENT / VALIDATION: Test comparison layout/upgrade prompts and require AOV lift without disproportionate conversion loss.

EVIDENCE ANCHORS: [R8] [R10]

Basket & Average Order Value Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

33. Use Cart-Side Cross-Sells With Strict Relevance

RAG TAGS: aov_down | items_per_order_down | cart | cross_sell | checkout_guardrail

TRIGGER: Basket size is weak but cart-to-checkout conversion is otherwise healthy.

RECOMMENDED ACTION: Offer one or two high-relevance add-ons in cart without obstructing the primary checkout action.

EXECUTION: Use cart contents to select accessories/refills, keep recommendations visually secondary, preserve checkout CTA prominence, and avoid forcing customers back into product browsing.

USE WHEN: Cross-sell relevance is high and checkout is already low-friction.

AVOID / DE-PRIORITIZE WHEN: High checkout abandonment, mobile clutter, or poor recommendation quality.

MONITOR: Attach rate, items/order, AOV, cart-to-checkout conversion, margin.

EFFICIENCY: Impact: Medium | Effort: Low-Medium | Speed: Fast | Confidence: Medium-High.

EXPERIMENT / VALIDATION: A/B test cart recommendation module; stop if checkout conversion falls more than incremental basket value compensates.

EVIDENCE ANCHORS: [R8] [R3] [R12]

34. Use Post-Purchase Upsell Where Checkout Simplicity Is More Valuable

RAG TAGS: aov_down | upsell | post_purchase | checkout | basket

TRIGGER: There is a strong add-on opportunity, but adding choices before payment would increase checkout complexity.

RECOMMENDED ACTION: Present a highly relevant add-on immediately after purchase or in confirmation flow instead of crowding checkout.

EXECUTION: Use the purchased item to select one compatible add-on, keep acceptance simple, state any shipping/fulfillment implications clearly, and avoid delaying confirmation of the original order.

USE WHEN: The platform can support post-purchase modification safely and the add-on is strongly related.

AVOID / DE-PRIORITIZE WHEN: Operational systems cannot reliably merge/ship added items, or customers need extensive comparison before purchase.

MONITOR: Post-purchase attach rate, AOV/revenue per order, fulfillment errors, cancellations, support contacts.

EFFICIENCY: Impact: Medium | Effort: Medium | Speed: Medium | Confidence: Medium.

EXPERIMENT / VALIDATION: Compare post-purchase offer vs no offer and monitor service/fulfillment guardrails.

EVIDENCE ANCHORS: [R8] [R12]

Pricing & Promotion Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

35. Target Discounts to the Problem Segment

RAG TAGS: promotion | discount | retention | acquisition | margin

TRIGGER: A price incentive may help, but the KPI problem is concentrated in a specific customer or product segment.

RECOMMENDED ACTION: Apply discounts only to the segment whose behavior needs to change—new buyers, lapsed customers, selected categories or threshold baskets.

EXECUTION: Define eligibility, hold out a control group, set discount and margin limits, suppress already-converting customers, and remove the offer after the objective is achieved.

USE WHEN: The segment and desired behavior are known and can be targeted.

AVOID / DE-PRIORITIZE WHEN: Site-wide discounting when average item price is already falling or the cause is non-price-related.

MONITOR: Incremental conversion/reactivation, average item price, AOV, contribution margin, cannibalization.

EFFICIENCY: Impact: Medium-High | Effort: Low-Medium | Speed: Fast | Confidence: High when targeted and measured.

EXPERIMENT / VALIDATION: Always use a holdout or control; judge on incremental profit/revenue, not redemption rate alone.

EVIDENCE ANCHORS: [R12] [R7]

36. Reduce Discount Depth Before Raising List Price

RAG TAGS: average_item_price_down | discounting | pricing | margin | promotion

TRIGGER: Average item price is falling and the decline is linked to heavier promotional depth or a larger share of discounted orders.

RECOMMENDED ACTION: Test shallower, more targeted promotions before making broad list-price increases.

EXECUTION: Measure effective discount by product/segment, identify orders where deep discounts do not materially improve conversion, narrow eligibility, and replace some price cuts with value-added offers such as bundles or threshold benefits.

USE WHEN: Promotion/discount data explains the price decline.

AVOID / DE-PRIORITIZE WHEN: Do not infer discounting as the cause if product mix shifted toward cheaper SKUs.

MONITOR: Average item price, promo share, conversion, contribution margin, units sold.

EFFICIENCY: Impact: High | Effort: Medium | Speed: Medium | Confidence: High when discount-depth data confirms cause.

EXPERIMENT / VALIDATION: Randomize discount depth in eligible cohorts with clear margin and conversion guardrails.

EVIDENCE ANCHORS: [R12] [R7]

Pricing & Promotion Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

37. Match Promotion Type to the Diagnosed KPI

RAG TAGS: promotion | decision_rule | aov | retention | acquisition

TRIGGER: The business is considering a promotion but needs to choose the correct mechanic for the diagnosed cause.

RECOMMENDED ACTION: Use first-order offers for acquisition, win-back offers for lapse, threshold rewards for AOV, bundle economics for items/order, and loyalty rewards for repeat behavior.

EXECUTION: Start from the KPI that must change, select one mechanic that directly affects that behavior, cap cost, and define the success metric before launch.

USE WHEN: A promotion is justified but the business wants to avoid generic blanket discounting.

AVOID / DE-PRIORITIZE WHEN: When the root cause is service, availability, seller capacity, payment or checkout friction; fix the root cause first.

MONITOR: The exact target KPI plus margin/cannibalization guardrails.

EFFICIENCY: Impact: High when well-matched | Effort: Low-Medium | Speed: Fast | Confidence: High conceptually, still requires testing.

EXPERIMENT / VALIDATION: Treat each promotion as a causal test with a control; stop mechanics that move the wrong KPI or destroy margin.

EVIDENCE ANCHORS: [R7] [R8] [R12]

38. Build Price-Pack Architecture Instead of One Flat Offer

RAG TAGS: average_item_price_down | pricing | pack_size | segmentation | aov

TRIGGER: Average item price is weak because customers concentrate in entry price points, while some customers may value larger packs or better variants.

RECOMMENDED ACTION: Create clear entry, standard and premium/large-pack choices that let customers self-select value rather than forcing a single price point.

EXECUTION: Map price/feature tiers, ensure meaningful differentiation, keep unit economics transparent, and merchandise the default tier deliberately based on customer value and conversion.

USE WHEN: Products support pack/feature variation and margins are known.

AVOID / DE-PRIORITIZE WHEN: Artificial tiering with no real customer benefit, confusing option overload or unreliable premium supply.

MONITOR: Average item price, mix by tier, AOV, conversion, margin/order.

EFFICIENCY: Impact: Medium | Effort: Medium | Speed: Medium | Confidence: Medium.

EXPERIMENT / VALIDATION: Test tier presentation and defaults; monitor whether premium mix rises without excessive conversion loss.

EVIDENCE ANCHORS: [R8] [R10] [R12]

Pricing & Promotion Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

39. Recover Premium Mix Through Merchandising, Not Forced Price Hikes

RAG TAGS: average_item_price_down | premium_mix | merchandising | pricing | product_mix

TRIGGER: Average item price declines because the product mix shifts toward lower-priced items rather than because individual prices changed.

RECOMMENDED ACTION: Improve discovery and comparison of higher-value products whose benefits justify the price.

EXECUTION: Identify premium SKUs with healthy conversion/returns, improve placement, comparison content and recommendations, and ensure the premium assortment is in stock and clearly differentiated.

USE WHEN: Mix analysis confirms a shift to cheaper products and premium products still have credible demand/value.

AVOID / DE-PRIORITIZE WHEN: Do not hide affordable products or manipulate ranking solely to raise price; customer relevance and conversion remain guardrails.

MONITOR: Average item price, premium mix, conversion, return rate, margin/order.

EFFICIENCY: Impact: Medium | Effort: Medium | Speed: Medium | Confidence: High when mix shift is measured.

EXPERIMENT / VALIDATION: Test premium merchandising on relevant traffic segments, not site-wide.

EVIDENCE ANCHORS: [R1] [R10] [R12]

40. Run a Priority-SKU Listing Quality Program

RAG TAGS: product | listing_quality | conversion | discoverability | seller_productivity

TRIGGER: Units/orders are weak for products with demand potential but incomplete or low-quality content.

RECOMMENDED ACTION: Fix listing quality on the SKUs that matter most before attempting broad catalog cleanup.

EXECUTION: Score products on title completeness, images, attributes, identifiers, category, description, price/availability consistency and seller information. Prioritize high-demand/high-margin or strategically important SKUs.

USE WHEN: Catalog quality varies materially and performance is concentrated in a manageable subset of SKUs.

AVOID / DE-PRIORITIZE WHEN: Do not spend equal effort on long-tail products with negligible demand while top SKUs remain broken.

MONITOR: Listing completeness score, conversion, product visibility, units sold, returns/questions.

EFFICIENCY: Impact: High | Effort: Medium | Speed: Medium | Confidence: High for obvious quality defects.

EXPERIMENT / VALIDATION: Use before/after matched-SKU comparisons and staggered remediation waves.

EVIDENCE ANCHORS: [R1] [R10]

Product & Merchandising Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

41. Make Availability and Price Data Accurate Everywhere

RAG TAGS: product | availability | price_consistency | conversion | trust

TRIGGER: Customers encounter out-of-stock products, stale availability, price mismatches or unavailable variants after clicking through.

RECOMMENDED ACTION: Synchronize price and availability across feed, marketplace/listing, product page and checkout.

EXECUTION: Increase update frequency for fast-changing inventory, expose explicit sold-out/preorder status, avoid redirecting unavailable products to unrelated pages, and preserve consistent variant/pricing information.

USE WHEN: Mismatch incidents, stockouts or feed errors are material.

AVOID / DE-PRIORITIZE WHEN: Do not use messaging alone to mask poor inventory accuracy; fix the data pipeline/source of truth.

MONITOR: Availability mismatch rate, stockout views, conversion, cancellations, feed disapprovals, customer complaints.

EFFICIENCY: Impact: High | Effort: Medium-High | Speed: Medium | Confidence: High.

EXPERIMENT / VALIDATION: Operational fix rather than classic A/B test; validate with discrepancy audits before/after rollout.

EVIDENCE ANCHORS: [R1] [R2]

42. Close Assortment Gaps Where Demand Exists

RAG TAGS: units_down | assortment | seller_acquisition | product_gap | marketplace

TRIGGER: Customer demand remains healthy but the catalog lacks important categories, brands, price tiers, sizes, colors or variants.

RECOMMENDED ACTION: Expand assortment only in demonstrated demand gaps, recruiting products/sellers that fill specific missing choices.

EXECUTION: Use search, browse, lost-sale, substitute, category and seller data to identify unmet demand; rank gaps by expected demand, margin and supply feasibility; onboard targeted assortment.

USE WHEN: Evidence indicates customers want products the marketplace/store does not currently offer reliably.

AVOID / DE-PRIORITIZE WHEN: Do not expand long-tail assortment simply to increase SKU count; avoid categories with weak demand or operational complexity the business cannot support.

MONITOR: Gap fill rate, conversion in target categories, units sold, seller/product activation, stockout/substitution rate.

EFFICIENCY: Impact: High | Effort: High | Speed: Medium-Long | Confidence: High with demand-gap evidence.

EXPERIMENT / VALIDATION: Pilot assortment expansion in the highest-confidence gap and measure incremental category sales.

EVIDENCE ANCHORS: [R11] [R14]

Product & Merchandising Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

43. Re-rank and Merchandise Around Relevance, Availability and Quality

RAG TAGS: product | ranking | merchandising | conversion | units_sold

TRIGGER: Customers browse a large assortment but low-quality, unavailable or poorly converting products receive too much visibility.

RECOMMENDED ACTION: Adjust merchandising/ranking so relevant, available, trustworthy products receive appropriate exposure.

EXECUTION: Use relevance plus availability, conversion quality, delivery reliability and returns as ranking/merchandising inputs. Create category collections for common missions and avoid over-concentrating exposure on a few sellers unless justified.

USE WHEN: The business controls on-site sorting/recommendation or marketplace exposure.

AVOID / DE-PRIORITIZE WHEN: Do not optimize only for past sales because that can entrench incumbents and hide new high-quality supply.

MONITOR: Category conversion, click-to-order, units sold, seller concentration, return/cancel rate.

EFFICIENCY: Impact: High | Effort: High | Speed: Medium | Confidence: Medium-High.

EXPERIMENT / VALIDATION: Run ranking experiments with guardrails for conversion, diversity, seller concentration and service quality.

EVIDENCE ANCHORS: [R11] [R12]

44. Improve Variant Completeness and Selection

RAG TAGS: product | variant | conversion | availability | catalog

TRIGGER: Products lose conversion because key sizes, colors, configurations or variant details are missing or hard to select.

RECOMMENDED ACTION: Complete variant data, group related options correctly and make the selected variant explicit through the acquisition-to-checkout path.

EXECUTION: Audit top products for missing variant attributes, correct parent/child grouping, ensure each variant has price/availability/image data, and preselect the promoted variant when users arrive from a listing or ad.

USE WHEN: Variant-heavy categories show option-related drop-off, returns or product mismatches.

AVOID / DE-PRIORITIZE WHEN: Do not create artificial variants that confuse customers or fragment reviews/inventory.

MONITOR: Variant availability, conversion, wrong-variant returns, selection errors, stockouts.

EFFICIENCY: Impact: Medium-High | Effort: Medium | Speed: Medium | Confidence: High for variant defects.

EXPERIMENT / VALIDATION: Fix high-volume variants first and compare conversion/return reasons against similar products.

EVIDENCE ANCHORS: [R2] [R1] [R10]

Seller Marketplace Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

45. Reactivate Inactive Sellers Before Recruiting More

RAG TAGS: active_sellers_down | seller_reactivation | marketplace | supply | efficiency

TRIGGER: Total seller count is stable but active sellers decline, or previously productive sellers stop listing/fulfilling.

RECOMMENDED ACTION: Diagnose and reactivate viable inactive sellers before spending heavily on new seller acquisition.

EXECUTION: Segment inactive sellers by prior sales, category importance and reason for inactivity. Resolve listing, inventory, payout, policy, fulfillment or support blockers; use targeted outreach for high-value sellers.

USE WHEN: Seller history identifies inactive but previously valuable supply.

AVOID / DE-PRIORITIZE WHEN: Do not reactivate chronically poor-quality, noncompliant or unprofitable sellers solely to raise seller count.

MONITOR: Active seller rate, reactivation rate, orders/seller, units/seller, cancellation/late-shipment rate.

EFFICIENCY: Impact: High | Effort: Medium | Speed: Medium | Confidence: High when inactive valuable sellers exist.

EXPERIMENT / VALIDATION: Run structured seller-recovery cohorts by root cause and track incremental active supply.

EVIDENCE ANCHORS: [R11]

46. Reduce Seller Time-to-Activation

RAG TAGS: new_sellers_down | seller_onboarding | marketplace | activation | supply

TRIGGER: New seller sign-ups or recruitment exist but too few become active sellers with live, purchasable listings.

RECOMMENDED ACTION: Simplify onboarding around the minimum steps required to publish a compliant, in-stock offer and fulfill the first order.

EXECUTION: Map the onboarding funnel, identify document/listing/inventory/fulfillment bottlenecks, provide clear checklists and assisted onboarding for priority sellers, and measure time from registration to first live SKU and first order.

USE WHEN: There is a seller pipeline and activation drop-off can be measured.

AVOID / DE-PRIORITIZE WHEN: Do not remove compliance, identity or quality controls; simplify guidance and workflow, not essential safeguards.

MONITOR: Seller activation rate, time to first listing/order, onboarding abandonment, early seller quality.

EFFICIENCY: Impact: High | Effort: Medium-High | Speed: Medium | Confidence: High with funnel data.

EXPERIMENT / VALIDATION: Test assisted onboarding or simplified workflows for a seller cohort.

EVIDENCE ANCHORS: [R14] [R11]

Seller Marketplace Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

47. Recruit Sellers Only for Verified Supply Gaps

RAG TAGS: seller_count_down | new_sellers_down | assortment | marketplace | targeted_recruitment

TRIGGER: Seller or assortment supply is inadequate in categories/regions where customer demand is demonstrably healthy.

RECOMMENDED ACTION: Target seller acquisition to specific demand gaps instead of maximizing total seller count.

EXECUTION: Rank categories/regions by unmet demand, low active-seller coverage, stockouts and service capacity. Define ideal seller profile, recruit specifically, and ensure onboarding support exists for the targeted assortment.

USE WHEN: Demand-side evidence supports the gap and the business can attribute seller coverage by category/region.

AVOID / DE-PRIORITIZE WHEN: Sellers are already increasing while orders/seller is falling; indiscriminate recruitment may dilute demand.

MONITOR: Active sellers in target gap, assortment coverage, conversion, orders/seller, stockout rate.

EFFICIENCY: Impact: High | Effort: High | Speed: Medium-Long | Confidence: High with gap evidence.

EXPERIMENT / VALIDATION: Recruit in one or two highest-priority gaps and compare category conversion/availability before broader rollout.

EVIDENCE ANCHORS: [R11] [R14]

48. Create a Seller Listing-Quality Improvement Program

RAG TAGS: orders_per_seller_down | seller_productivity | listing_quality | marketplace | conversion

TRIGGER: Active sellers remain present but orders per seller or units per seller decline because listings are incomplete, hard to discover or poor at converting.

RECOMMENDED ACTION: Give sellers concrete listing-quality standards and prioritized remediation tasks.

EXECUTION: Score title, images, attributes, identifiers, price/availability accuracy and product detail completeness; show sellers the highest-impact fixes; provide templates/tools and escalate chronic quality issues.

USE WHEN: Seller performance correlates with listing-quality gaps or catalog defects are observable.

AVOID / DE-PRIORITIZE WHEN: Do not blame sellers for low demand caused by platform traffic, category decline or oversupply; separate listing quality from demand allocation.

MONITOR: Orders/seller, units/seller, listing score, conversion, inactive listings, returns/questions.

EFFICIENCY: Impact: Medium-High | Effort: Medium | Speed: Medium | Confidence: High for measurable content defects.

EXPERIMENT / VALIDATION: Remediate a matched seller/SKU cohort and compare productivity changes.

EVIDENCE ANCHORS: [R10] [R11] [R1]

Seller Marketplace Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

49. Manage Seller Productivity With Service and Fulfillment Guardrails

RAG TAGS: orders_per_seller_down | units_per_seller_down | seller_performance | fulfillment | marketplace

TRIGGER: Seller productivity is weak and performance varies materially by cancellations, shipping, service or availability.

RECOMMENDED ACTION: Prioritize reliable sellers and coach or constrain sellers whose operational quality suppresses conversion and repeat demand.

EXECUTION: Track seller-level cancellation, late shipment, stock accuracy, returns and customer feedback. Provide dashboards/alerts, recovery plans and clear standards; improve exposure for consistently reliable supply where appropriate.

USE WHEN: Seller-level operational metrics exist and service quality explains performance differences.

AVOID / DE-PRIORITIZE WHEN: Do not use seller penalties when poor performance originates from platform/system failures outside seller control.

MONITOR: Orders/seller, units/seller, cancellation, on-time shipment/delivery proxy, seller feedback, repeat purchase.

EFFICIENCY: Impact: High | Effort: Medium-High | Speed: Medium | Confidence: High when service variation is clear.

EXPERIMENT / VALIDATION: Use phased coaching/ranking changes with quality and seller-concentration guardrails.

EVIDENCE ANCHORS: [R11]

50. Reduce Delivery-Time Variability, Not Just Average Delivery Time

RAG TAGS: retention | fulfillment | delivery | service | returning_customers_down

TRIGGER: Repeat purchasing, satisfaction or conversion is weak in segments with unreliable delivery estimates or high late-delivery variation.

RECOMMENDED ACTION: Improve predictability by identifying lanes/sellers/categories with unstable fulfillment and fixing the largest sources of variance.

EXECUTION: Measure promised vs actual delivery by seller/region/category, isolate long-tail delays, adjust promises where necessary, improve seller/fulfillment routing, and communicate exceptions early.

USE WHEN: Delivery data shows meaningful variability or late-order concentration.

AVOID / DE-PRIORITIZE WHEN: Do not promise faster dates simply to improve conversion if operations cannot reliably meet them.

MONITOR: Late-delivery rate, delivery variance, cancellations, support contacts, repeat purchase, seller performance.

EFFICIENCY: Impact: High | Effort: High | Speed: Medium-Long | Confidence: High when lateness is linked to customer outcomes.

EXPERIMENT / VALIDATION: Operational pilots by lane/seller; compare repeat and cancellation metrics before/after while controlling seasonality.

EVIDENCE ANCHORS: [R11]

Fulfillment & Service Playbook

Retrieve by trigger/cause. Each card is intentionally self-contained for chunking and vector search.

51. Build Cancellation and Stockout Recovery Paths

RAG TAGS: units_down | cancellation | stockout | availability | recovery

TRIGGER: Orders/units are lost through cancellations, unavailable products or failed seller fulfillment.

RECOMMENDED ACTION: Recover demand by preventing avoidable cancellations and offering relevant substitutes or fast alternatives when fulfillment fails.

EXECUTION: Classify cancellation causes, fix inventory accuracy for preventable cases, identify substitute products/sellers, notify customers quickly, preserve price/value where reasonable, and track whether recovered demand converts.

USE WHEN: Cancellation/stockout data is available and lost demand is material.

AVOID / DE-PRIORITIZE WHEN: Do not substitute materially different products without clear customer choice; do not hide availability problems.

MONITOR: Cancellation rate, stockout rate, recovered orders, conversion on substitutes, support contacts, repeat purchase.

EFFICIENCY: Impact: High | Effort: Medium-High | Speed: Medium | Confidence: High for preventable cancellations.

EXPERIMENT / VALIDATION: Pilot substitute/recovery flows on selected categories and measure net recovered revenue plus satisfaction guardrails.

EVIDENCE ANCHORS: [R1] [R2] [R11]

52. Turn Returns and Support Data Into Corrective Actions

RAG TAGS: retention | returns | support | product_quality | seller_quality

TRIGGER: Returning customers, conversion or seller productivity deteriorate alongside rising returns, complaints or support contacts.

RECOMMENDED ACTION: Use return/support reasons as operational signals to fix product information, quality, seller behavior or fulfillment rather than treating returns only as a cost center.

EXECUTION: Standardize reason codes, rank issues by frequency and value, connect them to SKU/seller/category, correct the upstream cause, and improve resolution speed for affected customers.

USE WHEN: Return/support reason data is sufficiently structured to identify patterns.

AVOID / DE-PRIORITIZE WHEN: Do not suppress legitimate returns to improve the metric; optimize the cause and customer experience.

MONITOR: Return rate by reason, repeat purchase after return, support resolution time, seller/SKU defect rates, conversion.

EFFICIENCY: Impact: High | Effort: Medium-High | Speed: Medium | Confidence: High with reliable reason coding.

EXPERIMENT / VALIDATION: Use before/after remediation cohorts for specific defect reasons and track both returns and repeat behavior.

EVIDENCE ANCHORS: [R6] [R11]

Recommendation Prioritization & Cross-KPI Decision Rules

Use this after retrieval to choose what Agent 3 should recommend first.

A. Priority rule: Impact × Confidence × Speed ÷ Effort

Score each dimension 1–3. Prefer actions that directly address the confirmed cause, can move the KPI quickly, and do not require large irreversible investment before evidence exists.

Class	Interpretation	Default Agent 3 behavior	Typical example
Quick Win	High/medium impact, low effort, fast	Recommend first if cause-fit is strong	Win-back segment, guest checkout, cart recovery
Strategic	High impact, high effort	Recommend with phased rollout	Seller onboarding redesign, ranking changes, assortment expansion
Experiment	Uncertain impact but testable	Recommend as controlled test	Discount depth, premium merchandising, new threshold
Investigate	Cause evidence incomplete	Do not present as confirmed solution	Possible marketing issue without channel data

B. Cross-KPI rules that prevent generic recommendations

- Revenue ↓ + active customers ↓ + revenue/customer stable → prioritize customer acquisition/retention based on new vs returning driver; do not prioritize AOV.
- Revenue ↓ + customers stable + AOV ↓ + items/order ↓ → prioritize basket-building (cross-sell, bundles, thresholds); do not start with acquisition.
- Revenue ↓ + customers stable + orders/customer ↓ + AOV stable → prioritize purchase-frequency/retention actions.
- AOV ↓ + items/order stable + average item price ↓ → investigate discount depth or product mix; do not recommend bundles as the primary fix.
- New customers stable + returning customers ↓ → retention/recovery is primary; broad acquisition is secondary.
- Seller count stable + active sellers ↓ → seller reactivation/activation; do not recruit blindly.
- Sellers ↑ + orders/seller ↓ → diagnose demand dilution, ranking or seller productivity before adding more sellers.
- Units sold ↓ + availability/cancellations worsen → fix supply/fulfillment before marketing demand.
- Conversion/checkout deterioration + AOV problem → fix checkout leakage before adding basket complexity.
- Low cause confidence → recommend a cheap validation step before a costly intervention.

Output discipline: Agent 3 should normally return one primary recommendation, not a list of ten equal ideas. Supporting actions must explain why they are secondary and what evidence would cause reprioritization.

Evidence Anchors & Source Notes

The playbook translates common e-commerce operating practices into conditional action cards. Sources support the intervention patterns; individual businesses must validate causal impact in their own data.

[R1] Google Merchant Center Help — Tips to optimize your product data

<https://support.google.com/merchants/answer/7380908?hl=en>

[R2] Google Merchant Center Help — About landing page requirements

<https://support.google.com/merchants/answer/4752265?hl=en>

[R3] Baymard Institute — How to Reduce Cart Abandonment / Checkout Usability Research

<https://baymard.com/learn/reduce-cart-abandonment>

[R4] Stripe — E-commerce checkout best practices

<https://stripe.com/in/resources/more/ecommerce-checkout-best-practices>

[R5] Shopify — How To Use Retention Emails To Drive Repeat Sales (2026)

<https://www.shopify.com/in/blog/retention-email>

[R6] Shopify — Earning Repeat Customers: 7 Ways to Lift Retail CLV (2026)

<https://www.shopify.com/blog/strategies-to-turn-one-time-shoppers-into-repeat-customers>

[R7] Shopify — How to Offer Free Shipping for Ecommerce

<https://www.shopify.com/enterprise/blog/ecommerce-fulfillment-free-shipping>

[R8] Shopify — Cross Sell vs Upsell: Differences and Examples (2026)

<https://www.shopify.com/in/blog/upselling-and-cross-selling>

[R9] Google Merchant Center Help — Product Ratings basics

<https://support.google.com/merchants/answer/14620705?hl=en>

[R10] Amazon — How to create Amazon product listings (2026)

<https://sell.amazon.com/blog/amazon-product-listings>

[R11] Amazon — Seller Central: manage listings, inventory, performance and growth

<https://sell.amazon.com/tools/seller-central>

[R12] Optimizely — Web Experimentation / A/B testing workflow

<https://support.optimizely.com/hc/en-us/articles/4410289104013-Steps-to-create-an-experiment>

[R13] Google Merchant Center Help — Google Customer Reviews

<https://support.google.com/merchants/answer/14628991?hl=en>

[R14] Amazon — New Seller Guide / seller growth services

<https://sell.amazon.com/grow/>

RAG ingestion recommendation

For production RAG, ingest each action card as a separate chunk with metadata: card_id, primary_kpi, driver, direction, problem_type, action_category, impact, effort, speed and required_evidence. Keep this DOCX as the human-readable master, but preserve the same card boundaries in Markdown/JSON used for embeddings.